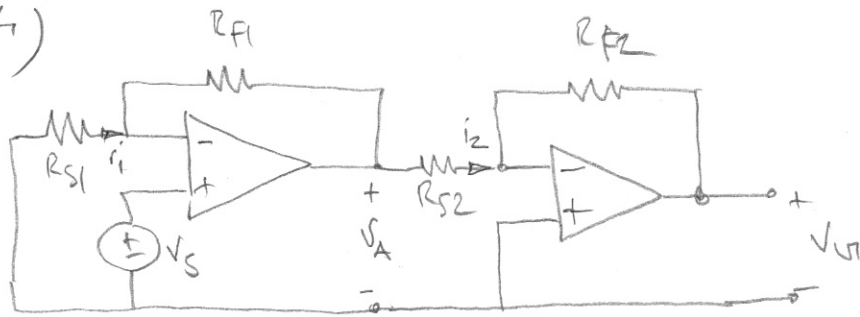


8.44)



$$R_{S1} = R_{S2} = 2.5 \text{ k}\Omega$$

$$R_{F1} = R_{F2} = 9 \text{ k}\Omega$$

Restwerte known  $\frac{V_U}{V_S}$  IDEAL OP-AMP.

Lösungsweg: Knotenstromanalyse, KIR P & MESS KIRPL

$$\begin{cases} + R_{S1} i_1 + V_S = 0 & (1) \\ - V_S + R_{F1} i_1 + V_A = 0 & (2) \\ - V_A + R_{S2} i_2 = 0 & (3) \\ - V_U - R_{F2} i_2 = 0 & (4) \end{cases}$$

$$(1) \text{ ger } i_1 = -\frac{V_S}{R_{S1}} \quad (5)$$

(5) i (2) ger

$$- V_S + R_{F1} \left( -\frac{V_S}{R_{S1}} \right) + V_A = 0$$

$$V_A = V_S \left( 1 + \frac{R_{F1}}{R_{S1}} \right) \quad (6)$$

(3) ger

$$i_2 = \frac{V_A}{R_{S2}}$$

(4) ger

$$V_U = - R_{F2} \frac{V_A}{R_{S2}} = - \frac{R_{F2}}{R_{S2}} \left( 1 + \frac{R_{F1}}{R_{S1}} \right) V_S \quad (6)$$

Numerisch:

$$V_U = - \frac{9}{2.5} \left( 1 + \frac{9}{2.5} \right) V_S = -8.2 V_S$$

$$\frac{V_U}{V_S} = -8.2 \text{ V}$$

$$\underline{\underline{S_{max}}} = \frac{V_U}{V_S} = -8.2 \text{ V}$$